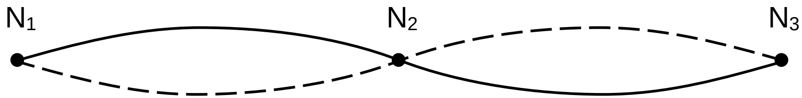
20 The diagram shows a standing wave on a string. The standing wave has three nodes N_1, N_2 and N_3.



Which statement is correct?

A All points on the string vibrate in phase.

B All points on the string vibrate with the same amplitude.

C Points equidistant from N_2 vibrate with the same frequency and in phase.

D Points equidistant from N_2 vibrate with the same frequency and the same amplitude.

Ans: D

All points between N_1 and N_2 vibrate in phase, and are all in antiphase with points between N_2 and N_3.

All points between a node and an anti-node vibrate with different amplitudes

All points vibrate with same frequency. Points to the left of a node vibrate in antiphase with the points to the right of a node.

21 A particle of charge –5.0 μC is projected from X towards Y with kinetic energy 250 μJ. When the